

Implementation of SGPA and CGPA Calculation Using Nested Loops & Decision-Making Constructs in C

The algorithm for calculating the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA) for 'm' semesters is given below. Each semester has 'n' courses, where 'n' may vary between semesters. Assume that the student has passed all courses. Prepare a test-case table covering valid, invalid, and boundary inputs; write the C program based on the given algorithm; compile and execute it; and test it against the prepared test cases.

Algorithm

1. Start.
2. Define the variables and their types: *global_id*, *m*, *n*, *semester*, *course*, *code*, *credit*, *GP*, *semester_TC*, *semester_TGC*, *cumulative_TC*, and *cumulative_TGC* as integers; *grade* as a character; and *SGPA* and *CGPA* as floating-point variables.
3. Get *global_id*.
4. Validate *global_id*: If *global_id* is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
5. Get *m*.
6. Validate *m*: If *m* is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
7. Set the cumulative values: $cumulative_TC = 0$, $cumulative_TGC = 0$
8. Repeat Steps 8.1 to 8.8 for *m* semesters:
 - 8.1. Set the semester totals:
 $semester_TC = 0$, $semester_TGC = 0$
 - 8.2. Get *n*.
 - 8.3. Validate *n*: If *n* is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
 - 8.4. Repeat Steps 8.4.1 to 8.4.9 for *n* courses:
 - 8.4.1. Get *code*.
 - 8.4.2. Validate *code*: If it is not positive, display an error message and request it again. Repeat until a positive value is entered.
 - 8.4.3. Get *credit*.
 - 8.4.4. Validate *credit*: If it is not positive, display an error message and request it again. Repeat until a positive value is entered.
 - 8.4.5. Get *grade*.
 - 8.4.6. Validate *grade*: Accept only O, A, B, C, D, E, or P. If the grade is invalid, display an error message and request it again. Repeat until a valid *grade* is entered.
 - 8.4.7. Find *GP* corresponding to *grade* using the university's conversion scale:
 $O = 10, A = 9, B = 8, C = 7, D = 6, E = 5, P = 4$
 - 8.4.8. Calculate the semester total grade credits:
 $semester_TGC = semester_TGC + (GP * credit)$
 - 8.4.9. Calculate the semester total credits:
 $semester_TC = semester_TC + credit$
 - 8.5. Calculate the *SGPA* of the current semester:
 $SGPA = semester_TGC / semester_TC$
 - 8.6. Display *semester* and *SGPA*.
 - 8.7. Add the semester grade credits to the cumulative grade credits:
 $cumulative_TGC = cumulative_TGC + semester_TGC$
 - 8.8. Add the semester credits to the cumulative credits:
 $cumulative_TC = cumulative_TC + semester_TC$
9. Calculate the *CGPA*:
 $CGPA = cumulative_TGC / cumulative_TC$
10. Display *global_id*, *m*, *cumulative_TC*, and *CGPA*.
11. Stop.